PROJECT 02 PDF READY

# Subscription Churn and Retention Radar

A must-have project for SaaS, telecom, app, and membership analytics roles.

Analyzed churn drivers, retention cohorts, and revenue-at-risk segments with SQL window functions and segmentation logic.

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## WHY IT MATTERS IN 2026

Retention work is still one of the best portfolio signals because it combines SQL, segmentation, revenue risk, and business storytelling.

## BUSINESS QUESTION

Which customer segments churn, when do they leave, and how much revenue is at risk?

## BUILD GOAL

Model churn in SQL, build a retention view, and score customers so the output feels useful to a growth or lifecycle team.

## SQL PATTERNS

- cohorts
- RFM segmentation
- LAG/LEAD
- retention curves
- risk bands
- revenue at risk

## TABLES TO CREATE

- customers
- subscriptions
- usage_events
- payments
- support_tickets
- churn_flags

## MANUAL

1. Define churn clearly first: cancellation, inactivity window, or contract expiry.
2. Build the schema and mark one table as the customer master for consistent joins.
3. Create cohort and retention queries by signup month, plan type, and tenure band.

4 Add a churn-risk score from usage, payment failures, complaints, and low engagement.

5 Summarize the highest-risk segments and attach one action per segment.

6 Export the final analysis tables as separate pages in a PDF report.

## DELIVERABLES

- schema and churn definition notes
- retention analysis SQL
- customer risk segmentation
- executive summary
- separate PDF report

## DIALECT NOTES

Use AGE, date_trunc, and interval comparisons in PostgreSQL for cohort and retention reporting.

## REFERENCES

GITHUB **ELDINBR / CUSTOMER-CHURN-SQL-ANALYSIS**
Direct SQL churn workflow with schema, exploration, churn analysis, segmentation, and revenue analysis.

GITHUB **VDD11 / CUSTOMER_CHURN_ANALYSIS**
Shows a star schema plus SQL analysis and a model layer, which is useful if you want a data-science edge.

YOUTUBE **CUSTOMER CHURN ANALYSIS PROJECT OVERVIEW**
A current project walkthrough that helps you frame churn as a portfolio story instead of a query exercise.